F - Plan Holidays

分值：525 分

Problem Statement

Takahashi is trying to decide his schedule for N days. Initially, none of the days are holidays.

高桥正规划 N 天的日程安排。初始时所有天都不是假期。

He can repeat either of the following operations any number of times:

他可以任意多次执行以下两种操作之一：

- Choose an integer i between 1 and N, inclusive, and make day i a holiday. This operation costs A_i.
- Choose an integer i between 2 and $N-1$, inclusive, such that both day $i-1$ and day $i+1$ are already holidays, and make day i a holiday. This operation is free.
- 选择一个介于 1 和 N 之间（含两端）的整数 i，将第 i 天设为假期。该操作花费 A_i。
- 选择一个介于 2 和 $N-1$ 之间（含两端）的整数 i，满足第 $i-1$ 天和第 $i+1$ 天均已为假期，将第 i 天设为假期。该操作免费。

Find the minimum total cost required to create a consecutive block of K or more holidays.

求构造出长度至少为 K 的连续假期段所需的最小总花费。

T test cases are given; solve each of them.

共有 T 组测试用例，你需要处理每一组数据。

Constraints

- $1 \le T \le 2 \times 10^5$
- $1 \le K \le N \le 2 \times 10^5$
- $1 \le A_i \le 10^9$
- All input values are integers.

- The sum of N over all test cases is at most 2×10^5.
- $1 \le T \le 2 \times 10^5$
- $1 \le K \le N \le 2 \times 10^5$
- $1 \le A_i \le 10^9$
- 所有输入值均为整数。
- 所有测试用例的 N 之和不超过 2×10^5。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

$$\begin{aligned}&T\\&\text{case}_1\\&\text{case}_2\\&\vdots\\&\text{case}_T\end{aligned}$$

Here, case_i denotes the input for the i-th test case. Each test case is given in the following format:

其中 case_i 表示第 i 个测试用例的输入内容。每个测试用例按照以下格式给出：

$$\begin{aligned}&N\ K\\&A_1\ A_2\ \ldots\ A_N\end{aligned}$$

Output

Output T lines. The i-th line should contain the answer for the i-th test case.

输出共 T 行，第 i 行应包含第 i 个测试用例的答案。

Sample Input 1

```
3
5 2
3 1 4 1 5
6 4
```

```
24 3 22 39 4 29
15 7
220651272 302798780 874479994 657822311 613294668 479624013 241168404 610547619 7625
```

Sample Output 1

```
2
29
1902064780
```

For the first test case, a consecutive block of at least two holidays can be created by performing operations as follows:

对于第一个测试用例，可以通过如下操作构造出长度至少为 2 的连续假期段：

- Make day 2 a holiday using the first type of operation. This costs 1.
- Make day 4 a holiday using the first type of operation. This costs 1.
- Make day 3 a holiday using the second type of operation. This is free.
- 使用第一种操作将第 2 天设为假期，花费 1。
- 使用第一种操作将第 4 天设为假期，花费 1。
- 使用第二种操作将第 3 天设为假期，免费。

The total cost of this sequence of operations is 2. It is impossible to create a consecutive block of two or more holidays with a cost less than 2, so output 2.

该操作序列的总花费为 2。不存在花费小于 2 的方案可以构造出长度至少为 2 的连续假期段，因此输出 2
。